

Year 11
Mathematics Extension 1
Assessment Task 3
Yearly Examination
2013

General Instructions

- Reading time – 5 minutes
- Working time – 1.5 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question

Note: Any time you have remaining should be spent revising your answers.

Total marks – 50

Section I

5 marks

- Attempt Questions 1 – 5

Section II

45 marks

- Attempt Questions 6, 7 & 8
- Start each question in a new writing booklet
- Write your name on the front cover of each booklet and writing sheet to be handed in.
- If you do not attempt a question, submit a blank booklet marked with your name and "N/A" on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Section I

5 marks

Attempt Questions 1 – 5

Use the multiple-choice answer sheet for Questions 1-5

1 How many solutions are there for $\cos \theta = -1$ in the domain $0^\circ \leq \theta \leq 720^\circ$?

- (A) 1
- (B) 5
- (C) 2
- (D) 4

2. Which of the following is an expression for $\frac{1}{(n-1)!} + \frac{1}{n!}$?

- (A) $\frac{n+1}{n!}$
- (B) $\frac{1}{(n-1)!}$
- (C) $\frac{n+1}{n(n-1)!}$
- (D) $\frac{2n-1}{n!(n-1)!}$

3. The acute angle between the lines $2x + y - 3 = 0$ and $x - 3y + 2 = 0$ is closest to

- (A) 18°
- (B) 45°
- (C) 67°
- (D) 82°

4. How many different arrangements of the letters of the word EXERCISE are possible?

- (A) 720
- (B) 6720
- (C) 20160
- (D) 40320

5. Which of the following is NOT an expression for $\cos 2\theta$?

- (A) $\cos^2 \theta - \sin^2 \theta$
- (B) $1 - 2\sin^2 \theta$
- (C) $\cos^2 \theta + \sin^2 \theta$
- (D) $2\cos^2 \theta - 1$

END OF SECTION I

Section II

45 marks

Attempt Questions 6 – 8

Answer each question in the appropriate writing booklet.

All necessary working should be shown in every question.

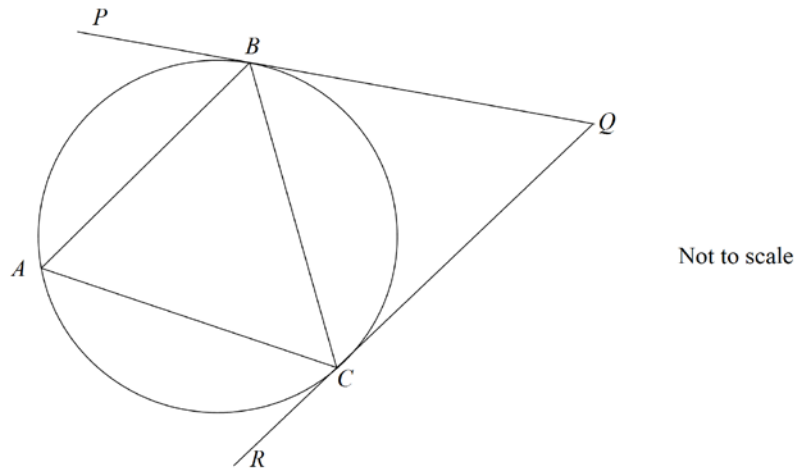
Question 6 (15 Marks)

Marks

- | | | |
|------|---|----------|
| (a) | Solve $\frac{2x}{6-x} \geq 1$ | 2 |
| (b) | Find the coordinates of the point P which divides the interval AB with end points $A(2,3)$ and $B(5,-7)$ in the ratio $4:9$. | 2 |
| (c) | Find the number of ways in which 2 brothers and 4 sisters can line up in a queue: | |
| (i) | Without restriction | 1 |
| (ii) | So that 2 brothers are NOT next to each other | 2 |
| (d) | By expressing $\tan 15^\circ$ as $\tan(60^\circ - 45^\circ)$, show that, as an exact value: | 2 |
| | $\tan 15^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1}.$ | |
| (e) | (i) Show that $\sin 3x = 3\sin x - 4\sin^3 x$ | 2 |
| | (ii) Hence, solve $3\sin x - 4\sin^3 x = \frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$. | 2 |

Question 6 continues on the next page.

- (f) In the diagram below PQ and RQ are tangents to the circle touching the circle at B and C respectively. $\angle BAC = 65^\circ$.



Copy or trace the diagram into your answer book.

Show that $\angle BQC = 50^\circ$

2

END OF QUESTION 6

Question 7 (15 Marks) Use a SEPARATE writing booklet**Marks**

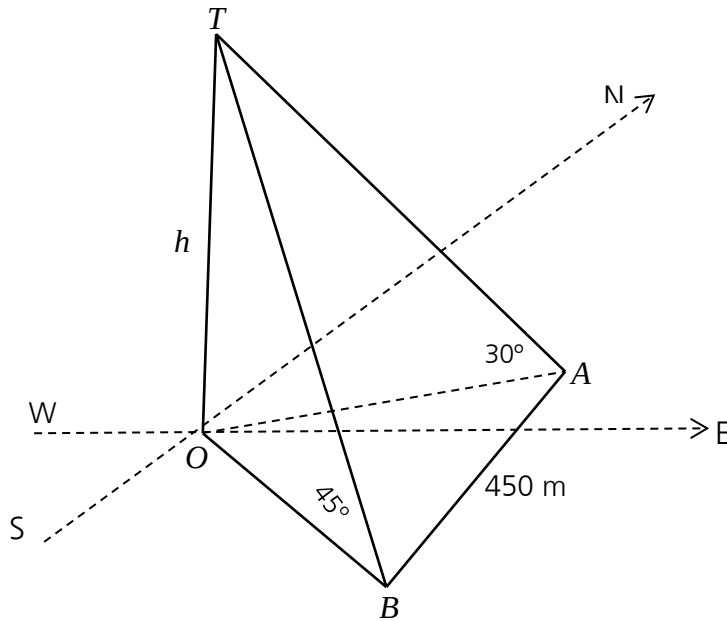
- (a) The curves $y = \sqrt{x}$ and $y = \frac{x^2}{4} - 2$ meet at the point (4,2). Calculate the acute angle between the tangents to curves at this point. **3**
- (b) By using $t = \tan \frac{\theta}{2}$, show that $\sin \theta = \frac{2t}{1+t^2}$ and $\cos \theta = \frac{1-t^2}{1+t^2}$ **3**
- (c) (i) Express $\sin x + \sqrt{3} \cos x$ in the form of $R \sin(x + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq 90^\circ$. **2**
- (ii) Hence or otherwise, find the general solution of $\sin x + \sqrt{3} \cos x = \sqrt{3}$. **2**
- (d) Show that $\tan^2 x - \tan^2 y = \frac{\cos^2 y - \cos^2 x}{\cos^2 x \cos^2 y}$ **2**
- (e) Solve $3^{2x} - 8 \times 3^x - 9 = 0$. **3**

END OF QUESTION 7

Question 8 (15 Marks) Use a SEPARATE writing booklet

Marks

(a)



A tower stands on level ground. An observation point A bears NE from the base of the tower. Another point of observation B , bears SE from the base of the tower A and B are 450 metres apart. The angle of elevation of the tower from A and B respectively are 30° and 45° .

(i) Show that $OB = \frac{h}{\tan 45^\circ}$ and $OA = \frac{h}{\tan 30^\circ}$. **1**

(ii) By using Pythagoras' theorem on $\triangle OAB$, find the value of h . **3**

(b) Let $f(x) = \frac{x}{x^2 - 1}$.

(i) For what values of x is $f(x)$ undefined? **1**

(ii) Show that $f'(x) < 0$ for all values of x for which the function is defined. **2**

(iii) Hence, sketch $y = f(x)$. Show all intercepts and asymptotes clearly. **2**

(iv) Find the equation of the tangent to the curve at $x = 2$. Express your answer in general form. **2**

Question 8 continues on the next page.

(c) Consider the circle $x^2 + y^2 - 2x - 14y + 25 = 0$.

(i) Show that if the line $y = mx$ intersects the circle in two distinct points, then $(1 + 7m)^2 - 25(1 + m^2) > 0$. **2**

(ii) For what values of m is the line $y = mx$ a tangent to the circle? **2**

END OF EXAMINATION

2013 Year 11 Yearly Examination

Student Name:

Mathematics Extension 1

Section I Multiple-Choice Answer Sheet

- | | | | | |
|----------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☐ |

2013 Year 11 Yearly Examination

Student Name: ...SOLWS...

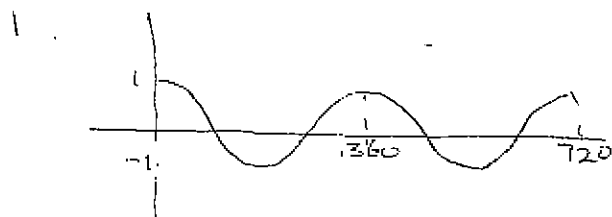
Mathematics Extension 1

Section I Multiple-Choice Answer Sheet

- | | | | | |
|---|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 | A ☐ | B ☐ | C ☒ | D ☐ |
| 2 | A ☒ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☒ |
| 4 | A ☐ | B ☒ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☒ | D ☐ |

2013 Prelim EI Yearly ATB.

MC sols:



sols $\cos \theta = -1$

(C) 2

2. $\frac{1}{(n-1)!} + \frac{1}{n!} = \frac{n+1}{n!}$ (A)

3. $2x + y - 3 = 0$ $x - 3y + 2 = 0$
 $y = 3 - 2x$ $y = \frac{1}{3}(x + 2)$
 $m_1 = -2$ $m_2 = \frac{1}{3}$

$$\tan \theta = \left| \frac{-2 - \frac{1}{3}}{1 + (-2)(\frac{1}{3})} \right|$$

$$= \left| -\frac{7}{1} \right|$$

$$= 7$$

$$\theta = 81^\circ 52'$$

(D)

4. EXERCISE $n = 8$ 3E

$$\# = \frac{8!}{3!} = 6720$$

(B)

5. (C)

Question 6

a) $\frac{2x}{6-x} > 1 \quad x \neq 6$

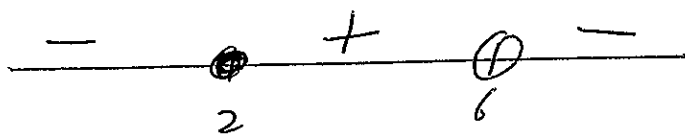
$$\left(\frac{2x}{6-x}\right)(6-x)^2 > (1)(6-x)^2$$

$$2x(6-x) > (6-x)^2$$

$$2x(6-x) - (6-x)^2 > 0$$

$$(6-x)[2x - (6-x)] > 0$$

$$(6-x)(3x-6) > 0$$



∴ solution is $2 \leq x < 6$

✓ (for any reasonable correct progress)

$x \in \mathbb{R}$ ✓ Both ineq. correct!

b) $A(2, 3) \quad B(5, -7) \quad m:n = 4:9$

$$x = \frac{mx_2 + nx_1}{m+n} = \frac{(4)(5) + (9)(2)}{13} = \frac{38}{13}$$

$$y = \frac{my_2 + ny_1}{m+n} = \frac{(4)(-7) + (9)(3)}{4+9} = -\frac{1}{13}$$

∴ $P\left(\frac{38}{13}, -\frac{1}{13}\right)$

✓✓
1 mark each
x and y values!

c) (i) Without restriction -

$$\text{no. of ways} = 6! \text{ or } 720$$

✓ correct

(ii) with 2 brothers next to each other -

$$\text{no. ways} = 2! \times 5! = 240$$

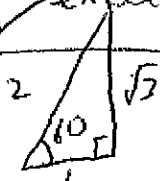
✓ 1 mark
2! × 5!

∴ 2 brothers not next to each other -

$$\begin{aligned} \text{no. of ways} &= 720 - 240 \\ &= \underline{\underline{480}} \end{aligned}$$

✓ 1 correct

d) $\tan(60-45) = \frac{\tan 60^\circ - \tan 45^\circ}{1 + \tan 60^\circ \tan 45^\circ}$

correct expansion
✓


$$= \frac{\sqrt{3} - 1}{1 + (\sqrt{3})(1)}$$

$$= \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

✓

e) (i) $\sin 3x = \sin(2x + x)$

correct expansion

$$\begin{aligned} &= \sin 2x \cos x + \cos 2x \sin x \\ &= 2 \sin x \cos x \cdot \cos x + \sin x (1 - 2 \sin^2 x) \\ &= 2 \sin x \cos^2 x + \sin x - 2 \sin^3 x \\ &= 2 \sin x (1 - \sin^2 x) + \sin x - 2 \sin^3 x \\ &= 2 \sin x - 2 \sin^3 x + \sin x - 2 \sin^3 x \\ &= 3 \sin x - 4 \sin^3 x \\ &= \text{RHS} \end{aligned}$$

* could use $\sin(4x - x)$.

✓

$$e) 3 \sin x - 4 \sin^3 x = \frac{1}{2}$$

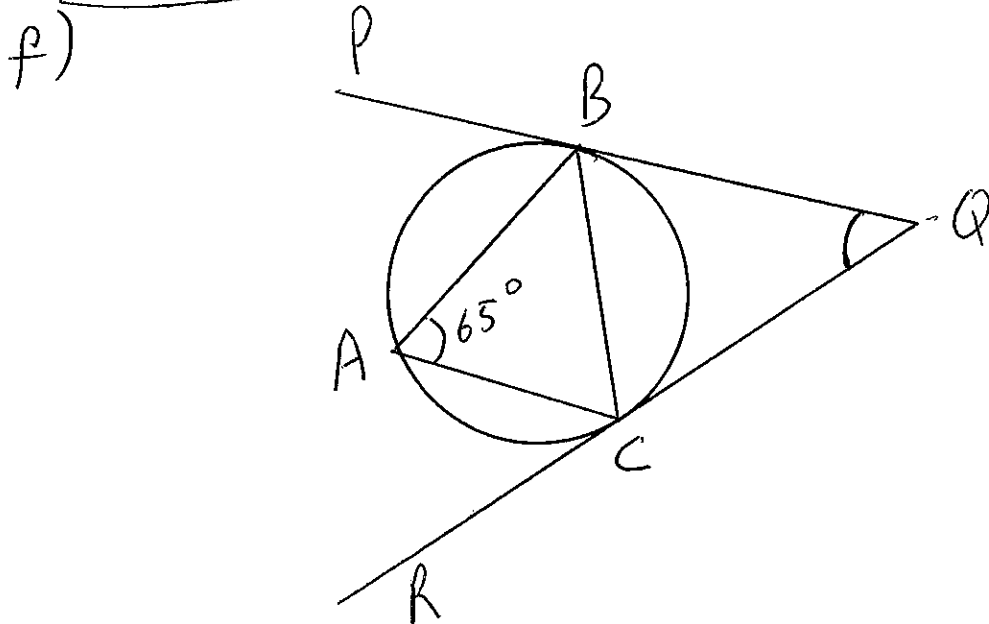
$$\sin 3x = \frac{1}{2}$$

1 mark for ✓
3x (change in domain)

1 mark ALL solns - ✓

$$3x = 30^\circ, 150^\circ, 390^\circ, 510^\circ, 750^\circ, 870^\circ$$

$$\therefore x = 10^\circ, 50^\circ, 130^\circ, 170^\circ, 250^\circ, 290^\circ$$



✓ $\angle BCQ = \angle BAC = 65^\circ$ (Angles in alternate segment)

✓ $BQ = QC$ (Equal tangents from external point.)

$\therefore \triangle BQC$ is isosceles

$$\therefore \angle CBQ = \angle QCB = 65^\circ$$

$$\therefore \angle BQC = 180 - 65 - 65 \text{ (Angle sum of } \triangle)$$

$$= 50^\circ$$

or/ 2 marks any correct proof

Question 7

a) $y = \sqrt{x} = x^{\frac{1}{2}}$

$$\frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

$$x=4 \Rightarrow m_1 = \frac{1}{2(\sqrt{4})} = \frac{1}{4}$$

✓ ~~(Finding m_1 & m_2)~~

$$y = \frac{x^2}{4} - 2 = \frac{1}{4}x^2 - 2$$

$$\frac{dy}{dx} = \frac{1}{2}x$$

$$x=4 \Rightarrow m_2 = \frac{1}{2}(4) = 2$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{2 - \frac{1}{4}}{1 + (2)(\frac{1}{4})} \right| = \frac{7}{6}$$

✓ correct use of formula.

$$\therefore \theta = \tan^{-1}\left(\frac{7}{6}\right)$$

$$= 49.3987^\circ$$

$$= \underline{49.40^\circ} \text{ or } \underline{49^\circ 24'}$$

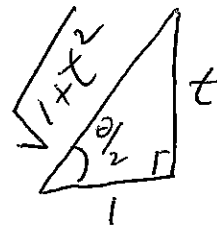
✓ correct

Question 7

$$b) \tan\left(\frac{\theta}{2}\right) = t = \frac{t}{1}$$

$$\sin\left(\frac{\theta}{2}\right) = \frac{t}{\sqrt{1+t^2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{1+t^2}}$$



✓ i mark both $\sin\frac{\theta}{2}$ and $\cos\frac{\theta}{2}$.

$$\begin{aligned} \sin \theta &= \sin 2\left(\frac{\theta}{2}\right) = 2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) \\ &= 2 \left(\frac{t}{\sqrt{1+t^2}}\right) \left(\frac{1}{\sqrt{1+t^2}}\right) \\ &= \frac{2t}{1+t^2} \quad \boxed{\checkmark \sin \theta} \end{aligned}$$

$$\begin{aligned} \text{and } \cos \theta &= \cos 2\left(\frac{\theta}{2}\right) = 1 - 2 \sin^2\left(\frac{\theta}{2}\right) \\ &= 1 - 2 \left(\frac{t}{\sqrt{1+t^2}}\right)^2 \\ &= 1 - \frac{2t^2}{1+t^2} \\ &= \frac{1+t^2-2t^2}{1+t^2} \\ &= \frac{1-t^2}{1+t^2} \quad \boxed{\checkmark \cos \theta} \end{aligned}$$

Question 7

c) (i) $\sqrt{3} \cos x + \sin x = R \sin(x + \alpha)$

$$R^2 = (\sqrt{3})^2 + 1^2 = 4$$

$$\underline{R = 2}$$

$$\therefore \sqrt{3} \cos x + \sin x = 2 \sin(x + \alpha)$$

$$\sqrt{3} \cos x + \sin x = 2 \sin x \cos \alpha + 2 \cos x \sin \alpha$$

$$\therefore \sqrt{3} \cos x = 2 \cancel{\cos x} \sin \alpha$$

$$\sin \alpha = \frac{\sqrt{3}}{2} \quad \text{--- (1)}$$

$$\text{and } \cancel{\sin x} = 2 \cancel{\sin x} \cos \alpha$$

$$\cos \alpha = \frac{1}{2} \quad \text{--- (2)}$$

From (1) and (2) α is in 1st quadrant

$$\alpha = \cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$$

$$\therefore \sqrt{3} \cos x + \sin x = \underline{2 \sin(x + 60^\circ)}$$

1 mark
 $R=2$
1 mark
 $\alpha=60^\circ$

(ii) $\sqrt{3} \cos x + \sin x = 2 \sin(x + 60^\circ) = \sqrt{3}$

$$\sin(x + 60^\circ) = \frac{\sqrt{3}}{2}$$

$$x + 60^\circ = 60^\circ, 120^\circ, \cancel{420^\circ}, \cancel{480^\circ}, \dots$$

$$x + 60^\circ = 60^\circ + 2\pi n \quad \left. \begin{matrix} \\ \\ \end{matrix} \right\} n \in \mathbb{Z} = 180n + (-1)^n 60^\circ$$

$$= 120^\circ + 2\pi n \quad \left. \begin{matrix} \\ \\ \end{matrix} \right\} \therefore x = 180n + (-1)^n 60^\circ - 60^\circ$$

$$x = 2\pi n, 60^\circ + 2\pi n$$

OR $x = 360n$ and

$$x = 360n \pm 60$$

Question 7

d) Show $\tan^2 x - \tan^2 y = \frac{\cos^2 y - \cos^2 x}{\cos^2 x \cos^2 y}$

$$RHS = \frac{\cos^2 y - \cos^2 x}{\cos^2 x \cos^2 y}$$

$$= \frac{\cancel{\cos^2 y}}{\cos^2 x \cancel{\cos^2 y}} - \frac{\cancel{\cos^2 x}}{\cancel{\cos^2 x} \cos^2 y}$$

$$= \frac{1}{\cos^2 x} - \frac{1}{\cos^2 y}$$

$$= \sec^2 x - \sec^2 y$$

$$= (\tan^2 x + 1) - (\tan^2 y + 1)$$

$$= \tan^2 x + 1 - \tan^2 y - 1$$

$$= \tan^2 x - \tan^2 y$$

$$= LHS$$

✓
correct progress

✓
obtaining result

e) $3^{2x} - 8 \cdot 3^x - 9 = 0$

Let $u = 3^x$

$$\therefore u^2 - 8u - 9 = 0$$

$$(u-9)(u+1) = 0$$

$$u = 9$$

ie; $3^x = 9$

$$\therefore x = 2$$

$$\therefore x = 2$$

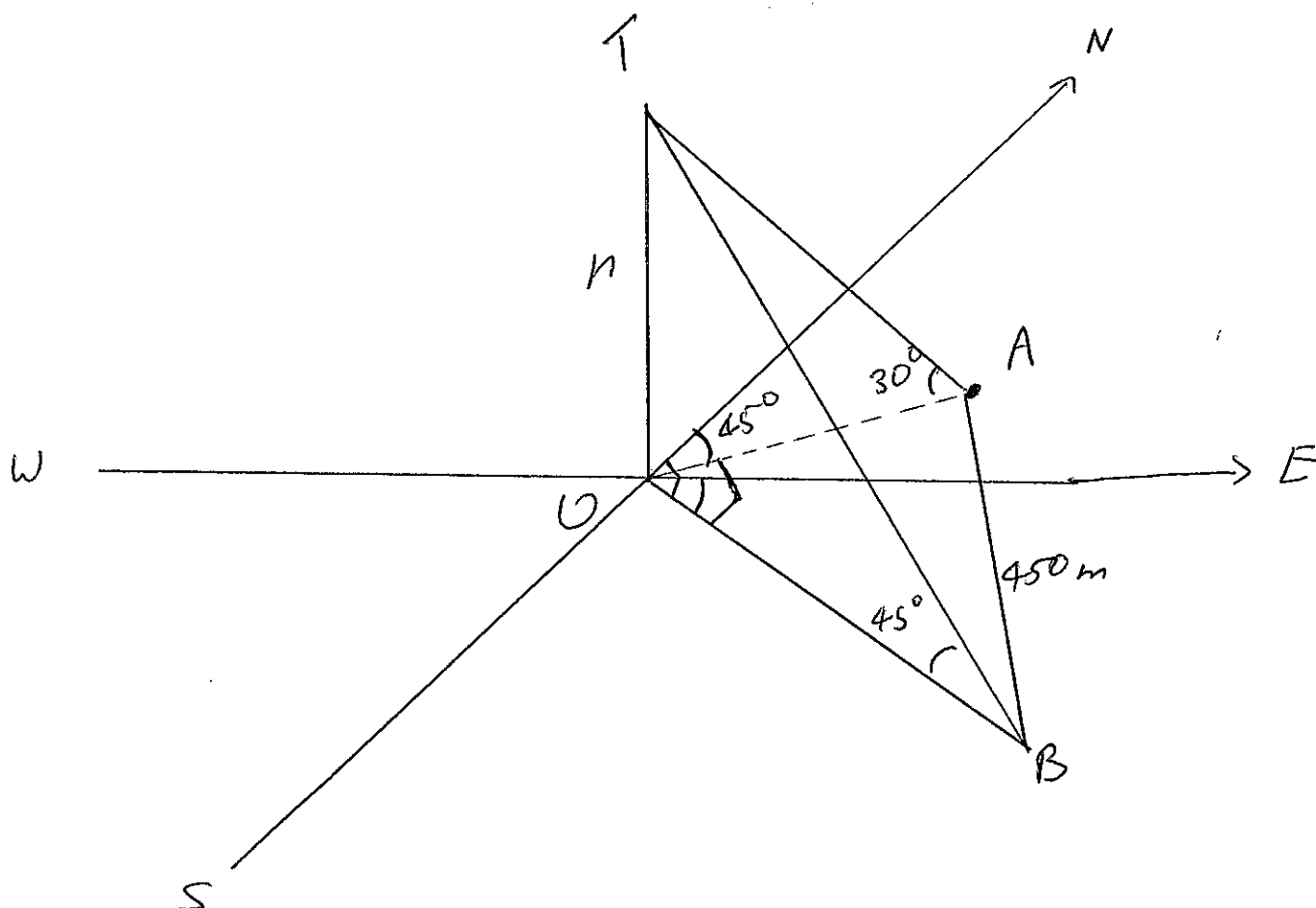
✓
correct quadratic

or $u = -1$
 $3^x = -1$ quadratic solns

x not defined

is the solution ✓
for $x=2$
ONLY

Question 8



$$(i) \tan 45 = \frac{h}{OB} \Rightarrow OB = \frac{h}{\tan 45}$$

$$\tan 30 = \frac{h}{OA} \Rightarrow OA = \frac{h}{\tan 30}$$

✓
(for both correct)

$$(ii) OA^2 + OB^2 = 450^2$$

$$\left(\frac{h}{\tan 45}\right)^2 + \left(\frac{h}{\tan 30}\right)^2 = 202500$$

$$h^2 + \frac{h^2}{\left(\frac{1}{\sqrt{3}}\right)^2} = 202500 \quad (\tan 30 = \frac{1}{\sqrt{3}})$$

$$h^2 + 3h^2 = 202500$$

$$4h^2 = 202500$$

$$\underline{h = 225 \text{ m}}$$

✓
correct use
Pythag.

✓
correct
subs.

✓
correct

Question 8

b) $f(x) = \frac{x}{x^2-1} \quad x \neq \pm 1$

(i) $x = \pm 1$ $f(x)$ undefined

✓ 1 mark
BOTH

(ii) $f(x) = \frac{x}{x^2-1}$

using $f\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

$$f'(x) = \frac{(1)(x^2-1) - (x)(2x)}{(x^2-1)^2}$$

✓ correct
use Quotient
Rule

$$= \frac{x^2 - 1 - 2x^2}{(x^2-1)^2}$$

$$= \frac{-x^2 - 1}{(x^2-1)^2}$$

$$\therefore f'(x) = \frac{-(x^2+1)}{(x^2-1)^2}$$

Since $-(x^2+1)$ is < 0 for all x

and $(x^2-1)^2 > 0$ for all x

✓
correct
explanation

$$\therefore f'(x) = \frac{-(x^2+1)}{(x^2-1)^2} < 0 \text{ for all } x$$

Question 8

$$(iii) f(x) = \frac{x}{x^2 - 1}$$

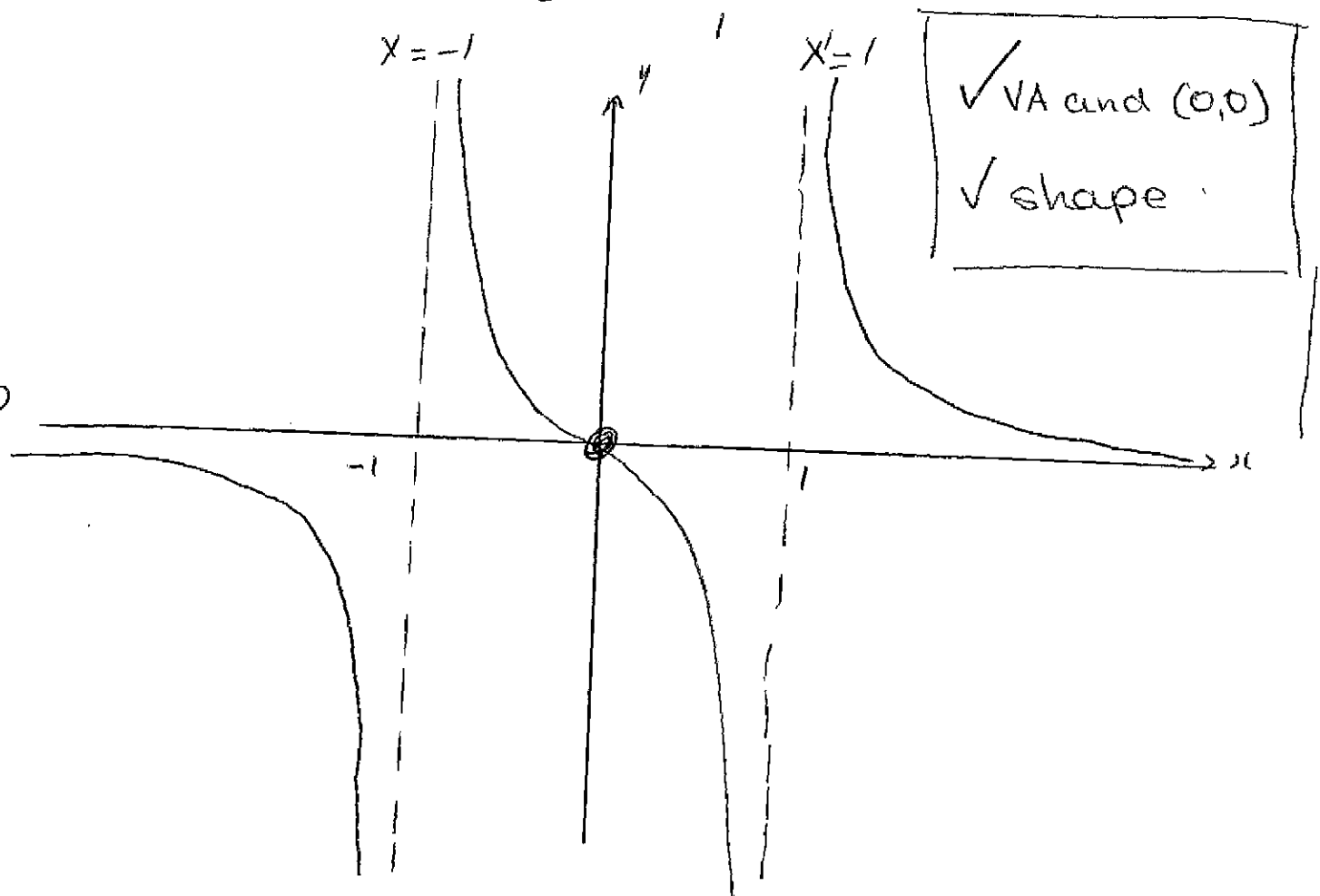
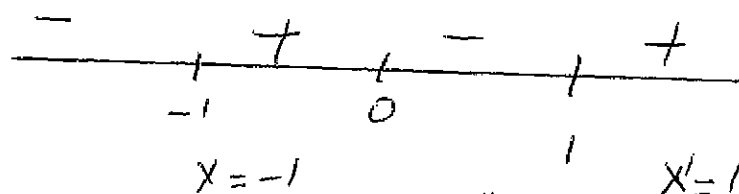
$x=0 \Rightarrow y=0$ $(0,0)$ is y -intercept

$x=1$ and $x=-1$ are vertical asymptotes.

horizontal asymptote

$$y = \lim_{x \rightarrow \infty} \frac{x}{x^2 - 1} = 0$$

$y=0$ is an asymptote



Question 8

$$(iv) x=2 \Rightarrow f(x) = \frac{2}{2^2-1} = \frac{2}{3}$$

Tangent at $(2, \frac{2}{3})$

$$x=2 \Rightarrow m = \frac{dy}{dx} = -\frac{(2^2+1)}{(2^2-1)^2} = -\frac{5}{9}$$

$\therefore$ equation of tangent:

$$y - \frac{2}{3} = -\frac{5}{9}(x-2)$$

$$9y - 6 = -5x + 10$$

$$\underline{5x + 9y - 16 = 0}$$

✓
correct use
of formula.

✓
must be in
general form.

(c) Solving $x^2 + y^2 - 2x - 14y + 25 = 0$ and
 $y = mx$ simultaneously:

$$x^2 + (mx)^2 - 2x - 14mx + 25 = 0$$

$$(1+m^2)x^2 - x(2+14m) + 25 = 0$$

For distinct roots $b^2 - 4ac > 0$

$$(2+14m)^2 - (4)(25)(1+m^2) > 0$$

$$4(1+7m)^2 - 4(25)(1+m^2) > 0$$

$\div$ by 4 both sides:

$$\therefore \underline{(1+7m)^2 - 25(1+m^2) > 0}$$

✓
correct
sub for
pt of inter-
section.

✓
correct
expression

Question 8

(c) (ii) For $y = mx$ to be a tangent

$$b^2 - 4ac = 0$$

$$0 = (1+m^2)^2 - 25(1+m^2)$$

$$1 + 14m + 49m^2 - 25 - 25m^2 = 0$$

$$24m^2 + 14m - 24 = 0$$

$$12m^2 + 7m - 12 = 0$$

$$12m^2 + 16m - 9m - 12 = 0$$

$$4m(3m+4) - 3(3m+4) = 0$$

$$(3m+4)(4m-3) = 0$$

$$\therefore m = -\frac{4}{3} \text{ or } m = \frac{3}{4}$$



✓
Both
correct
values.

for setting
 $\Delta = 0$